❶ Randomly generate the initial training data.

❷ Record incumbent value and retrain model.

❸ Find the point maximizing the acquisition score, query the objective function, and update the training data.

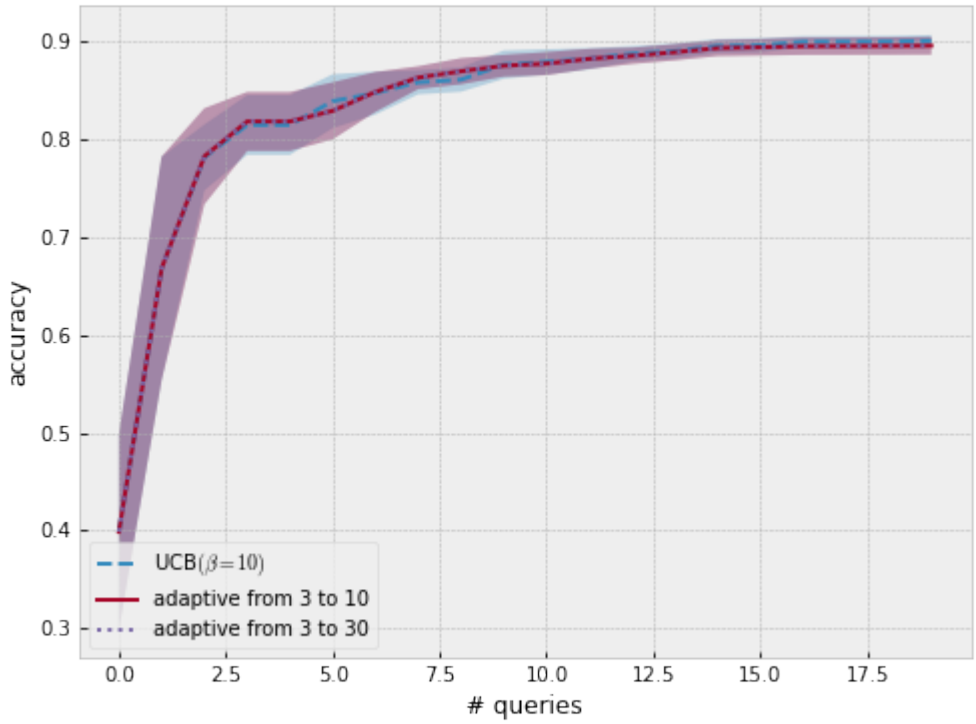


Figure A.12 Progress made by two adaptive versions of the Upper Confidence Bound policy. The policy is robust against the end value of the tradeoff parameter.

Figure A.12 shows the optimization performance of the two adaptive versions against the best performing fixed value $= 10$. These versions are comparable, and changing the end value from 10 to 30 doesn't affect optimization performance too much.

4. Run the Thompson sampling policy and observe its aggregated performance.

We implement Thompson sampling as follows:

```
num_candidates = 2000
num_repeats = 10

incumbents = torch.zeros((num_repeats, num_queries))

for trial in range(num_repeats):
    ... ❶

    for i in tqdm(range(num_queries)):
        ... ❷

        sobol = torch.quasirandom.SobolEngine(1, scramble=True)
        candidate_x = sobol.draw(num_candidates)
        candidate_x = bounds[0] + (bounds[1] - bounds[0]) * candidate_x

        ts = botorch.generation.MaxPosteriorSampling(model, replacement=False)
        next_x = ts(candidate_x, num_samples=1)

        ... ❸
```

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